

Course outline

SFWRENG 2AA4 Software Design I: Introduction to Software Development (2026 Winter)

INSTRUCTORS

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GENERAL INFORMATION

CALENDAR DESCRIPTION

Software life cycle, quality attributes, requirements documentation, specifying behavior; classes and objects, interface specification; creational, structural, and behavioral software design patterns; implementation in code, reviews, testing and verification.

LEARNING OBJECTIVES¹

This course introduces students to the profession of software engineering and the software development process. Students will gain an appreciation of software design strategies that facilitate working on large applications with multiple developers. Students will study the basic principles of software design. They will learn how to specify, design, implement, and verify software units in an object-oriented language like Java. It builds on the knowledge acquired in **SFWRENG 2DM3** (*Discrete Mathematics with Applications I*), **SFWRENG 2OP3** (*Object-oriented Programming*) and the experiential one from **SFWRENG 2XC3** (*Software Engineering Practice and Experience: Development Basics*). At the end of the course, students should:

- **Know and understand:**
 - The engineering principles associated with developing software.
 - The role and methods of design through systematic trade-offs.
 - The techniques of connecting design with implementation.
 - Modern software delivery methods.
- **Be able to:**
 - Systematically design software from small to large scale.
 - Seek design trade-offs by evaluating design alternatives.
 - Choose the right methods and principles for the software engineering process.
 - Carry out the software engineering process in a collaborative setting.

¹ Based on the descriptions of Dr. Spencer Smith's (2AA4 – Winter 2020) and Dr. Sebastien Mosser (2AA4 – Winter 2024).

COURSE PAGE

The course uses *Avenue2Learn* for pedagogical material.

SCHEDULE

Slot	Monday	Tuesday	Wednesday	Thursday	Friday
10:30AM 11:20 AM	T03				
11:30 AM 12:20 PM		Lecture		Lecture	Lecture
12:30 PM 1:20 PM		T02			T01
1:30 PM 2:20 PM					
2:30 PM 3:20 PM					
3:30 PM 4:20 PM					

Attendance of lectures and tutorials is not mandatory but highly recommended.

The Instructor reserves the right to make lectures and/or tutorials mandatory in case class performance necessitates. Such changes will be announced clearly and in a timely fashion.

The Instructor reserves the right to cancel a scheduled lecture according to unexpected circumstances.

TEXTBOOK AND COURSE MATERIAL

Slides from the lecture will be available on Avenue. (These are not lecture notes though.)

Students will be required to read chapters from the following book:

- Introduction to Software Design with Java (Martin Robillard, 2022)
 - <https://link.springer.com/book/10.1007/978-3-030-97899-0>
 - **Freely available** for McMaster students in PDF

ONLINE COMPONENT & COMMUNICATION

The course uses A2L as its main website to make lecture slides, tutorial material, and assignments available. To store program code, students will use GitHub.

COURSE CONTENTS

OUTLINE

The course will (tentatively) cover the following topics.

Week	Date	Topic	Module	Format
1	Jan 6	Welcome to 2AA4	Module 1: Basics	In person
	Jan 8	The principles of software <i>engineering</i>		In person
	Jan 9	The role of design and the impact of Generative AI		In person
2	Jan 13	Object-oriented design 1	Module 2: OO principles and design	In person
	Jan 15	Object-oriented design 2		In person
	Jan 16	UML 1		In person
3	Jan 20	UML 2		In person
	Jan 22	UML Practice		In person
	Jan 23	SOLID principles 1		In person
4	Jan 27	SOLID principles 2		In person
	Jan 29	STUPID principles		In person
	Jan 30	GRASP (Modular design)		In person
5	Feb 3	Types and interfaces		In person
	Feb 5	Composition		In person
	Feb 6	Testing		In person
6	Feb 10	Refactoring	Module 3: Change	In person
	Feb 12	MIDTERM		In person
	Feb 13	MIS and ADT		In person
7	Recess week – No lectures			
8	Feb 24	Design patterns: Intro	Module 4: Design patterns	In person
	Feb 26	Design patterns 1 (C): Singleton, Factory		Video
	Feb 27	Design patterns 2 (B1): Observer, Visitor		Video
9	Mar 3	Design patterns 2 (Principles): Equality, Uniqueness, Liskov, IoC, DI + DP		Video
	Mar 5	Design patterns 3 (S): Flyweight, Adapter, Decorator		Video
	Mar 6	Design patterns 3 (B2): Command, Chain of responsibility, Template method		Video
10	Mar 10	Q&A + design practice (potential flipped classroom)		In person
	Mar 12	Builder pattern. Advanced patterns: MVC, Fluent API		In person
	Mar 13	Q&A + design practice (potential flipped classroom)		In person
11	Mar 17	Code quality 1	Module 5: Code quality	In person
	Mar 19	Code quality 2		In person
	Mar 20	Architecture	Module 6: Advanced topics	In person
12	Mar 24	Delivery, deployment, maintenance, retirement		In person
	Mar 26	Modeling processes		In person
	Mar 27	Validation and verification		In person
13	Mar 31	Case study or guest lecture		In person
	Apr 2	Sustainable and green software		In person
	Apr 3	(Gen)AI in software engineering and design		In person
14	Apr 7	Wrap-up		In person

The instructor reserves the right to modify this tentative planning during the term. If any modification becomes necessary, reasonable notice and communication with the students will be given with an explanation and the opportunity to comment on changes.

GRADING & EVALUATION

ID	Type	Kind	Weight	Available	Due
Assignment 1	Assignment	Group	15%	Week 2	Week 5
Assignment 2	Assignment	Group	12.5%	Week 5	Week 8
Midterm test	Assessment	Individual	25%	Feb 12, 2026	
Assignment 3	Assignment	Group	12.5%	Week 8	Week 12
Final exam	Assessment	Individual	35%	TBA (Registrar)	

The instructor reserves the right to modify this plan during the term. Such changes will be announced clearly and in a timely fashion on A2L.

Assignments reports are submitted on A2L. Program code is submitted on GitHub.

Group assignments are expected to be executed by **groups of four students from the same tutorial section**.

Additional key details, tips, and recommendations for students are available in the *Assignment supplement* on A2L.

MSAF POLICY

Type-A MSAFs are automatically accepted when submitted with the following relief:

- **Assignments:** the MSAF gives the student **three (3) additional days** to finish their part in the project. The team the student is part of must clearly document the student's expected contributions as well as provide an assessment of the work distribution in the specific assignment.
- **Midterm:** the assessment is cancelled and **its weight is carried to the final exam** (i.e., the final exam is worth 60% of the final mark).
- **There is no need to contact the instructor for Type-A MSAFs.** Submit your MSAF through Mosaic.

Type-B MSAFs will be discussed with the student **after** being accepted by the Faculty.

ACADEMIC INTEGRITY & PLAGIARISM²

You are expected to exhibit honesty and ethical behavior in all aspects of the learning process. The academic credentials you earn are rooted in honesty and academic integrity principles. Assignments must be solved by the team only, **any** outside source (this includes asking other people or using any books or information found on the web) **must** be documented. In compliance with the senate regulations on academic integrity, I remind you that: **People who let other people copy are as guilty as the ones who copy**. You can consult outside sources, meaning textbooks or the web, but **any** use of an outside source **must** be documented. **Code plagiarism detection tools will be used systematically**.

Similarly, group assignments must be solved by members of that group only, and the University policies apply as well (see "Advisory Statements" below).

The use of generative AI is strictly forbidden until further notice and unless asked to be used in an assignment. If asked to use generative AI tools, the use of such tools is restricted to the scope specified in the assignment.

Suspicious cases will be investigated within an academic dishonesty procedure. We choose this avenue to provide every part with a fair and equitable opportunity to represent their case.

² Adapted from Dr. Jacques Carette outline for SFWRENG 3GB3 – Fall 2021

MAPPING TO GRADUATE ATTRIBUTES

Attribute	Indicator		Evaluation	
			Ai	Ex.
Knowledge Base	1.1	Competence in Mathematics	✓	
	1.4	Competence in specialized engineering knowledge	✓	✓
Problem analysis	2.1	Identifies and states reasonable assumptions and suitable engineering fundamentals, before proposing a solution path to a problem	✓	✓
Investigation	3.1	Selects appropriately from relevant knowledge base to plan appropriate data collection methods and analysis strategies	✓	
	3.2	Synthesizes the results of an investigation to reach valid conclusions	✓	
Design	4.1	Defines the problem by identifying relevant context, constraints, and prior approaches before exploring potential design solutions	✓	✓
	4.2	Explores a breadth of potential solutions, considering their benefits and trade-offs as they relate to the project requirements	✓	✓
	4.3	Develops models/prototypes; tests, evaluates, and iterates as appropriate	✓	
Use of engineering tools	5.2	Successfully uses engineering tools	✓	
Professionalism	8.1	Describes the duty of a Professional Engineer to the public, client, employer, and the profession	✓	✓

ADVISORY STATEMENTS

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: *"Grade of F assigned for academic dishonesty"*), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the **Academic Integrity Policy**, located at <https://secretariat.mcmaster.ca/university-policies-proceduresguidelines/>

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

AUTHENTICITY / PLAGIARISM

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

COURSES WITH AN ON-LINE COMPONENT

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

ONLINE PROCTORING

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s [Academic Accommodation of Students with Disabilities policy](#).

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar’s Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

COPYRIGHT AND RECORDING

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors. Distribution of course material is prohibited.

Recording of the instructors and distribution of such material is prohibited and **will be met with immediate legal action** from the Lead Instructor.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.